Proof. For an RV-valued f, put $h = \mathrm{vrv} \circ f$; for a Γ-valued f, put $h = f$. We retain the induction on k, with the case $k = 0$ immediate. For $k > 1$ and $t \in \mathrm{pr}_1(U)$, the inductive hypothesis makes $h(U_t)$ doubly bounded. Its finite bound

$$\gamma_t = \sup\{|h(t, x)| \mid x \in U_t\}$$

exists by definable completeness of Γ, and the least-upper-bound formula makes $t \longmapsto \gamma_t$ definable. The projection $\mathrm{pr}_1(U)$ is doubly bounded, so the unary Γ-valued case bounds all γ_t and hence bounds $h(U)$. Thus both claims reduce to the case $k = 1$ and a definable function $g : U \longrightarrow \Gamma$. Then, by C-minimality, we may assume that g vrv-contracts to a function $g_\downarrow : \mathrm{vrv}(U) \longrightarrow \Gamma$. Since definable functions in Γ are piecewise $\mathbb{Q}$-linear, the range of $g_\downarrow$ must be doubly bounded. □

Resolved referee R037 [Gap] – Uniformity in the bounded RV-image lemma.
Original source line 1017. The induction on k must control the bounds uniformly in the remaining coordinates; boundedness on each fiber alone does not give a bound on their union. A precise route is to consider the relation of possible values $\mathrm{vrv}(f(u))$ over $\mathrm{vrv}(u) \in \Gamma^k$. Residue-field/value-group orthogonality makes each such fiber finite, uniformly; stable embeddedness and elimination in divisible ordered abelian groups then give finitely many piecewise rational-affine branches. These have bounded images on a bounded box. Include this justification, or an equivalent explicit uniform induction.

Author resolution. The author retains the original induction. Its fiber bounds are chosen definably as finite suprema of the absolute output values; the unary Γ-valued case then bounds this definable bound function uniformly. The proof now spells out that step, without replacing the induction by a relation of all input and output values.

Definition 2.42. Unapplied R038 proposal for author review. Keep the same punctured-neighborhood limit condition, replace the redundant subfamily $L' \subseteq L$ by L, and allow the formula locally without a global range bound. A finite affine limit already forces local boundedness. Global boundedness remains explicit in results that require it, such as Lemma 2.45. The reused proof of uniqueness of the closure of a minimal finite-point limit set needs no boundedness hypothesis. No change to minimality or the puncture is proposed here.

Let $A \subseteq \mathrm{VF}^n$ and $f : A \longrightarrow \mathcal{P}(\mathrm{VF}^m)$ be a definable function. For $a \in \mathrm{VF}^n$ and $L \subseteq \mathrm{VF}^m$, we say that L is a *limit set of f at a*, written as $\lim_{A \to a} f \subseteq L$, if for every $\epsilon \in \Gamma$ there is a $\delta \in \Gamma$ such that if $c \in \mathfrak{o}(a, \delta) \cap (A \setminus \{a\})$ then

$$f(c) \subseteq \bigcup_{b \in L} \mathfrak{o}(b, \epsilon).$$

This is a local definition and imposes no global boundedness hypothesis. A finite limit set forces the union of the values $f(c)$ to be bounded on some punctured neighborhood of a. Whenever boundedness of the range is assumed, it means boundedness of $\bigcup_{c \in A} f(c)$.

A limit set L of f at a is *minimal* if no proper subset of L is a limit set of f at a.

Referee R038 [Clarification; revised] – Simplify the limit-set definition.
Original source line 1046. Say explicitly that $\bigcup_{a \in A} f(a)$ is bounded. The auxiliary subset L' does no work: the displayed condition is equivalent to $f(c) \subseteq \bigcup_{b \in L} \mathfrak{o}(b, \epsilon)$. The following definition of continuity is applied to finitary functions without a global boundedness hypothesis, so the present definition needs a compatible extension. One can use the same formula for arbitrary correspondences whenever a finite affine limit exists, or formulate limits in $(\mathbb{P}^1(\mathrm{VF}))^m$ from the outset; local boundedness near the point suffices for affine local uses.

Observe that if $\lim_{A \to a} f \subseteq L$ and $b \in L$ is not isolated in L then actually $\lim_{A \to a} f \subseteq L \setminus b$. So in a minimal limit set every element is isolated. Moreover, if a minimal limit set L exists then its topological closure is unique: